T1

October 26, 2026

1 Study Project: Adversarial Machine Learning

1.1 Part I: Statistical Analysis and Generation of Process Data

1.2 Group 15

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1.3.1 Task 1: Statistical Analysis of Physical Readings

The goal of this task is to analyse the properties of the WADI datasets, WADI_14days and WADI_attackdata.

- Compute K-S statistic and compare the sensor readings from datasets
- Count the set of system states and the percentages

1.3.2 Libray Import and Data Loading

[1]:
```
print(f"Start: Import Library and Load the datasets")
# Library Import
import pandas as pd
import numpy as np
from pathlib import Path
from collections import Counter

# Define the base directory path
base_path = Path.home() / "Projects/P1/data/"
cache_train_file = base_path / "processed/cache_train_file.parquet"
cache_test_file = base_path / "processed/cache_test_file.parquet"

# Load normal 14days data and attack data from CSV or Cache
# Load 14days Dataset
if cache_train_file.exists():
   print("Loading fast cached Parquet data-14days dataset")
   train17_df = pd.read_parquet(cache_train_file)
else:
   print("Cache not found. Load data from CSV file-14days dataset")
   # Skip 4 meta data rows
   train17_df = pd.read_csv(base_path / "raw/2017/WADI_14days.csv", skiprows=4)
   # Save the cache for next time
```

```
    train17_df.to_parquet(cache_train_file)

# Load attackdata Dataset
if cache_test_file.exists():
    print("Loading fast cached Parquet data-attackdata dataset")
    test17_df = pd.read_parquet(cache_test_file)
else:
    print("Cache not found. Load data from CSV file-attackdata dataset")
    test17_df = pd.read_csv(base_path / "raw/2017/WADI_attackdata.csv")
    # Save the cache for next time
    test17_df.to_parquet(cache_test_file)

# Remove the long prefix Column Name
prefix_to_remove = r"\\WIN-25J4RO10SBF\LOG_DATA\SUTD_WADI\LOG_DATA"

# Replace the string in all column names
train17_df.columns = train17_df.columns.str.replace(prefix_to_remove, "",␣
 ↪regex=False)
test17_df.columns = test17_df.columns.str.replace(prefix_to_remove, "",␣
 ↪regex=False)
```

```
Start: Import Library and Load the datasets
Loading fast cached Parquet data-14days dataset
Loading fast cached Parquet data-attackdata dataset
```

[2]:
```
print(f"Data shape of WADI_14days")
print(train17_df.shape)
```

```
Data shape of WADI_14days
(1209601, 130)
```

[3]:
```
print(f"Data Info of WADI_14days")
print(train17_df.info(verbose=True,show_counts=True))
print(train17_df)
```

```
Data Info of WADI_14days
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1209601 entries, 0 to 1209600
Data columns (total 130 columns):
 #    Column                      Non-Null Count    Dtype
---   ------                      --------------    -----
 0    Row                         1209601 non-null  int64
 1    Date                        1209601 non-null  object
 2    Time                        1209601 non-null  object
 3    \1_AIT_001_PV               1209601 non-null  float64
 4    \1_AIT_002_PV               1209589 non-null  float64
 5    \1_AIT_003_PV               1209601 non-null  float64
 6    \1_AIT_004_PV               1209595 non-null  float64
```

```
7    \1_AIT_005_PV            1209601 non-null  float64
8    \1_FIT_001_PV            1209601 non-null  float64
9    \1_LS_001_AL             1209601 non-null  int64
10   \1_LS_002_AL             1209601 non-null  int64
11   \1_LT_001_PV             1209601 non-null  float64
12   \1_MV_001_STATUS         1209601 non-null  int64
13   \1_MV_002_STATUS         1209601 non-null  int64
14   \1_MV_003_STATUS         1209601 non-null  int64
15   \1_MV_004_STATUS         1209601 non-null  int64
16   \1_P_001_STATUS          1209601 non-null  int64
17   \1_P_002_STATUS          1209601 non-null  int64
18   \1_P_003_STATUS          1209601 non-null  int64
19   \1_P_004_STATUS          1209601 non-null  int64
20   \1_P_005_STATUS          1209601 non-null  int64
21   \1_P_006_STATUS          1209601 non-null  int64
22   \2_DPIT_001_PV           1209601 non-null  float64
23   \2_FIC_101_CO            1209601 non-null  float64
24   \2_FIC_101_PV            1209601 non-null  float64
25   \2_FIC_101_SP            1209601 non-null  float64
26   \2_FIC_201_CO            1209601 non-null  float64
27   \2_FIC_201_PV            1209601 non-null  float64
28   \2_FIC_201_SP            1209601 non-null  float64
29   \2_FIC_301_CO            1209601 non-null  float64
30   \2_FIC_301_PV            1209601 non-null  float64
31   \2_FIC_301_SP            1209601 non-null  float64
32   \2_FIC_401_CO            1209601 non-null  float64
33   \2_FIC_401_PV            1209601 non-null  float64
34   \2_FIC_401_SP            1209601 non-null  float64
35   \2_FIC_501_CO            1209601 non-null  float64
36   \2_FIC_501_PV            1209601 non-null  float64
37   \2_FIC_501_SP            1209601 non-null  float64
38   \2_FIC_601_CO            1209601 non-null  float64
39   \2_FIC_601_PV            1209601 non-null  float64
40   \2_FIC_601_SP            1209601 non-null  float64
41   \2_FIT_001_PV            1209601 non-null  float64
42   \2_FIT_002_PV            1209601 non-null  float64
43   \2_FIT_003_PV            1209601 non-null  float64
44   \2_FQ_101_PV             1209601 non-null  float64
45   \2_FQ_201_PV             1209601 non-null  float64
46   \2_FQ_301_PV             1209601 non-null  float64
47   \2_FQ_401_PV             1209601 non-null  float64
48   \2_FQ_501_PV             1209601 non-null  float64
49   \2_FQ_601_PV             1209601 non-null  float64
50   \2_LS_001_AL             0 non-null        float64
51   \2_LS_002_AL             0 non-null        float64
52   \2_LS_101_AH             1209601 non-null  int64
53   \2_LS_101_AL             1209601 non-null  int64
54   \2_LS_201_AH             1209601 non-null  int64
```

```
55   \2_LS_201_AL             1209601 non-null  int64
56   \2_LS_301_AH             1209601 non-null  int64
57   \2_LS_301_AL             1209601 non-null  int64
58   \2_LS_401_AH             1209601 non-null  int64
59   \2_LS_401_AL             1209601 non-null  int64
60   \2_LS_501_AH             1209601 non-null  int64
61   \2_LS_501_AL             1209601 non-null  int64
62   \2_LS_601_AH             1209601 non-null  int64
63   \2_LS_601_AL             1209601 non-null  int64
64   \2_LT_001_PV             1209601 non-null  float64
65   \2_LT_002_PV             1209601 non-null  float64
66   \2_MCV_007_CO            1209601 non-null  int64
67   \2_MCV_101_CO            1209601 non-null  float64
68   \2_MCV_201_CO            1209601 non-null  float64
69   \2_MCV_301_CO            1209601 non-null  float64
70   \2_MCV_401_CO            1209601 non-null  float64
71   \2_MCV_501_CO            1209601 non-null  float64
72   \2_MCV_601_CO            1209601 non-null  float64
73   \2_MV_001_STATUS         1209601 non-null  int64
74   \2_MV_002_STATUS         1209601 non-null  int64
75   \2_MV_003_STATUS         1209601 non-null  int64
76   \2_MV_004_STATUS         1209601 non-null  int64
77   \2_MV_005_STATUS         1209601 non-null  int64
78   \2_MV_006_STATUS         1209601 non-null  int64
79   \2_MV_009_STATUS         1209601 non-null  int64
80   \2_MV_101_STATUS         1209601 non-null  int64
81   \2_MV_201_STATUS         1209601 non-null  int64
82   \2_MV_301_STATUS         1209601 non-null  int64
83   \2_MV_401_STATUS         1209601 non-null  int64
84   \2_MV_501_STATUS         1209601 non-null  int64
85   \2_MV_601_STATUS         1209601 non-null  int64
86   \2_P_001_STATUS          0 non-null        float64
87   \2_P_002_STATUS          0 non-null        float64
88   \2_P_003_SPEED           1209601 non-null  float64
89   \2_P_003_STATUS          1209601 non-null  int64
90   \2_P_004_SPEED           1209601 non-null  float64
91   \2_P_004_STATUS          1209601 non-null  int64
92   \2_PIC_003_CO            1209601 non-null  float64
93   \2_PIC_003_PV            1209601 non-null  float64
94   \2_PIC_003_SP            1209601 non-null  int64
95   \2_PIT_001_PV            1209601 non-null  float64
96   \2_PIT_002_PV            1209601 non-null  float64
97   \2_PIT_003_PV            1209601 non-null  float64
98   \2_SV_101_STATUS         1209601 non-null  int64
99   \2_SV_201_STATUS         1209601 non-null  int64
100  \2_SV_301_STATUS         1209601 non-null  int64
101  \2_SV_401_STATUS         1209601 non-null  int64
102  \2_SV_501_STATUS         1209601 non-null  int64
```

```
 103  \2_SV_601_STATUS           1209601 non-null  int64
 104  \2A_AIT_001_PV             1209601 non-null  float64
 105  \2A_AIT_002_PV             1209601 non-null  float64
 106  \2A_AIT_003_PV             1209601 non-null  float64
 107  \2A_AIT_004_PV             1209601 non-null  float64
 108  \2B_AIT_001_PV             1209601 non-null  float64
 109  \2B_AIT_002_PV             1209601 non-null  float64
 110  \2B_AIT_003_PV             1209601 non-null  float64
 111  \2B_AIT_004_PV             1209581 non-null  float64
 112  \3_AIT_001_PV              1209601 non-null  float64
 113  \3_AIT_002_PV              1209595 non-null  float64
 114  \3_AIT_003_PV              1209601 non-null  float64
 115  \3_AIT_004_PV              1209595 non-null  float64
 116  \3_AIT_005_PV              1209601 non-null  float64
 117  \3_FIT_001_PV              1209601 non-null  float64
 118  \3_LS_001_AL               1209601 non-null  int64
 119  \3_LT_001_PV               1209601 non-null  float64
 120  \3_MV_001_STATUS           1209601 non-null  int64
 121  \3_MV_002_STATUS           1209601 non-null  int64
 122  \3_MV_003_STATUS           1209601 non-null  int64
 123  \3_P_001_STATUS            1209601 non-null  int64
 124  \3_P_002_STATUS            1209601 non-null  int64
 125  \3_P_003_STATUS            1209601 non-null  int64
 126  \3_P_004_STATUS            1209601 non-null  int64
 127  \LEAK_DIFF_PRESSURE        1209601 non-null  float64
 128  \PLANT_START_STOP_LOG      1209601 non-null  int64
 129  \TOTAL_CONS_REQUIRED_FLOW  1209601 non-null  float64
dtypes: float64(71), int64(57), object(2)
memory usage: 1.2+ GB
None
             Row       Date            Time  \1_AIT_001_PV  \1_AIT_002_PV  \
0              1  9/25/2017  6:00:00.000 PM        171.155       0.619473
1              2  9/25/2017  6:00:01.000 PM        171.155       0.619473
2              3  9/25/2017  6:00:02.000 PM        171.155       0.619473
3              4  9/25/2017  6:00:03.000 PM        171.155       0.607477
4              5  9/25/2017  6:00:04.000 PM        171.155       0.607477
…            …         …               …              …             …
1209596  1209597  10/9/2017  5:59:56.000 PM        164.211       0.511488
1209597  1209598  10/9/2017  5:59:57.000 PM        164.211       0.511488
1209598  1209599  10/9/2017  5:59:58.000 PM        164.211       0.511488
1209599  1209600  10/9/2017  5:59:59.000 PM        164.211       0.511488
1209600  1209601  10/9/2017  6:00:00.000 PM        164.210       0.529486

         \1_AIT_003_PV  \1_AIT_004_PV  \1_AIT_005_PV  \1_FIT_001_PV  \
0              11.5759        504.645       0.318319       0.001157
1              11.5759        504.645       0.318319       0.001157
2              11.5759        504.645       0.318319       0.001157
3              11.5725        504.673       0.318438       0.001207
```

```
4              11.5725        504.673       0.318438       0.001207
…                …              …             …              …
1209596        11.9928        482.452       0.331198       0.001182
1209597        11.9928        482.452       0.331198       0.001182
1209598        11.9928        482.452       0.331198       0.001182
1209599        11.9928        482.452       0.331198       0.001182
1209600        11.9972        482.480       0.331167       0.001273

         \1_LS_001_AL  …  \3_MV_001_STATUS  \3_MV_002_STATUS  \
0                   0  …                 1                 1
1                   0  …                 1                 1
2                   0  …                 1                 1
3                   0  …                 1                 1
4                   0  …                 1                 1
…                 …  …                 …                 …
1209596             0  …                 1                 1
1209597             0  …                 1                 1
1209598             0  …                 1                 1
1209599             0  …                 1                 1
1209600             0  …                 1                 1

         \3_MV_003_STATUS  \3_P_001_STATUS  \3_P_002_STATUS  \3_P_003_STATUS  \
0                       1                 1                 1                 1
1                       1                 1                 1                 1
2                       1                 1                 1                 1
3                       1                 1                 1                 1
4                       1                 1                 1                 1
…                     …                 …                 …                 …
1209596                 1                 1                 1                 1
1209597                 1                 1                 1                 1
1209598                 1                 1                 1                 1
1209599                 1                 1                 1                 1
1209600                 1                 1                 1                 1

         \3_P_004_STATUS  \LEAK_DIFF_PRESSURE  \PLANT_START_STOP_LOG  \
0                       1              67.9651                      1
1                       1              67.9651                      1
2                       1              67.9651                      1
3                       1              67.1948                      1
4                       1              67.1948                      1
…                     …                …                          …
1209596                 1              62.8802                      1
1209597                 1              62.8802                      1
1209598                 1              62.8802                      1
1209599                 1              62.8802                      1
1209600                 1              62.6226                      1

         \TOTAL_CONS_REQUIRED_FLOW
```

```
0                              0.68
1                              0.68
2                              0.68
3                              0.68
4                              0.68
…                              …
1209596                        0.39
1209597                        0.39
1209598                        0.39
1209599                        0.39
1209600                        0.39

[1209601 rows x 130 columns]
```

[7]:
```
print(f"Data Shape of WADI_attackdata")
print(test17_df.shape)
```

```
Data Shape of WADI_attackdata
(172801, 130)
```

[8]:
```
print(f"Data Info of WADI_attackdata")
print(test17_df.info(verbose=True, show_counts=True))
print(test17_df)
```

```
Data Info of WADI_attackdata
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 172801 entries, 0 to 172800
Data columns (total 130 columns):
 #    Column                      Non-Null Count   Dtype
---   ------                      --------------   -----
 0    Row                         172801 non-null  int64
 1    Date                        172801 non-null  object
 2    Time                        172801 non-null  object
 3    \1_AIT_001_PV               172801 non-null  float64
 4    \1_AIT_002_PV               172801 non-null  float64
 5    \1_AIT_003_PV               172801 non-null  float64
 6    \1_AIT_004_PV               172801 non-null  float64
 7    \1_AIT_005_PV               172801 non-null  float64
 8    \1_FIT_001_PV               172801 non-null  float64
 9    \1_LS_001_AL                172801 non-null  int64
 10   \1_LS_002_AL                172801 non-null  int64
 11   \1_LT_001_PV                172801 non-null  float64
 12   \1_MV_001_STATUS            172801 non-null  int64
 13   \1_MV_002_STATUS            172801 non-null  int64
 14   \1_MV_003_STATUS            172801 non-null  int64
 15   \1_MV_004_STATUS            172801 non-null  int64
 16   \1_P_001_STATUS             172801 non-null  int64
 17   \1_P_002_STATUS             172801 non-null  int64
```

```
18   \1_P_003_STATUS           172801 non-null  int64
19   \1_P_004_STATUS           172801 non-null  int64
20   \1_P_005_STATUS           172801 non-null  int64
21   \1_P_006_STATUS           172801 non-null  int64
22   \2_DPIT_001_PV            172801 non-null  float64
23   \2_FIC_101_CO             172801 non-null  float64
24   \2_FIC_101_PV             172801 non-null  float64
25   \2_FIC_101_SP             172801 non-null  float64
26   \2_FIC_201_CO             172801 non-null  float64
27   \2_FIC_201_PV             172801 non-null  float64
28   \2_FIC_201_SP             172801 non-null  float64
29   \2_FIC_301_CO             172801 non-null  float64
30   \2_FIC_301_PV             172801 non-null  float64
31   \2_FIC_301_SP             172801 non-null  float64
32   \2_FIC_401_CO             172801 non-null  float64
33   \2_FIC_401_PV             172801 non-null  float64
34   \2_FIC_401_SP             172801 non-null  float64
35   \2_FIC_501_CO             172801 non-null  float64
36   \2_FIC_501_PV             172801 non-null  float64
37   \2_FIC_501_SP             172801 non-null  float64
38   \2_FIC_601_CO             172801 non-null  float64
39   \2_FIC_601_PV             172801 non-null  float64
40   \2_FIC_601_SP             172801 non-null  float64
41   \2_FIT_001_PV             172801 non-null  float64
42   \2_FIT_002_PV             172801 non-null  float64
43   \2_FIT_003_PV             172801 non-null  float64
44   \2_FQ_101_PV              172801 non-null  float64
45   \2_FQ_201_PV              172801 non-null  float64
46   \2_FQ_301_PV              172801 non-null  float64
47   \2_FQ_401_PV              172801 non-null  float64
48   \2_FQ_501_PV              172801 non-null  float64
49   \2_FQ_601_PV              172801 non-null  float64
50   \2_LS_001_AL              0 non-null       float64
51   \2_LS_002_AL              0 non-null       float64
52   \2_LS_101_AH              172801 non-null  int64
53   \2_LS_101_AL              172801 non-null  int64
54   \2_LS_201_AH              172801 non-null  int64
55   \2_LS_201_AL              172801 non-null  int64
56   \2_LS_301_AH              172801 non-null  int64
57   \2_LS_301_AL              172801 non-null  int64
58   \2_LS_401_AH              172801 non-null  int64
59   \2_LS_401_AL              172801 non-null  int64
60   \2_LS_501_AH              172801 non-null  int64
61   \2_LS_501_AL              172801 non-null  int64
62   \2_LS_601_AH              172801 non-null  int64
63   \2_LS_601_AL              172801 non-null  int64
64   \2_LT_001_PV              172801 non-null  float64
65   \2_LT_002_PV              172801 non-null  float64
```

```
66   \2_MCV_007_CO             172801 non-null  int64
67   \2_MCV_101_CO             172801 non-null  float64
68   \2_MCV_201_CO             172801 non-null  float64
69   \2_MCV_301_CO             172801 non-null  float64
70   \2_MCV_401_CO             172801 non-null  float64
71   \2_MCV_501_CO             172801 non-null  float64
72   \2_MCV_601_CO             172801 non-null  float64
73   \2_MV_001_STATUS          172801 non-null  int64
74   \2_MV_002_STATUS          172801 non-null  int64
75   \2_MV_003_STATUS          172801 non-null  int64
76   \2_MV_004_STATUS          172801 non-null  int64
77   \2_MV_005_STATUS          172801 non-null  int64
78   \2_MV_006_STATUS          172801 non-null  int64
79   \2_MV_009_STATUS          172801 non-null  int64
80   \2_MV_101_STATUS          172801 non-null  int64
81   \2_MV_201_STATUS          172801 non-null  int64
82   \2_MV_301_STATUS          172801 non-null  int64
83   \2_MV_401_STATUS          172801 non-null  int64
84   \2_MV_501_STATUS          172801 non-null  int64
85   \2_MV_601_STATUS          172801 non-null  int64
86   \2_P_001_STATUS           0 non-null       float64
87   \2_P_002_STATUS           0 non-null       float64
88   \2_P_003_SPEED            172801 non-null  float64
89   \2_P_003_STATUS           172801 non-null  int64
90   \2_P_004_SPEED            172801 non-null  float64
91   \2_P_004_STATUS           172801 non-null  int64
92   \2_PIC_003_CO             172801 non-null  float64
93   \2_PIC_003_PV             172801 non-null  float64
94   \2_PIC_003_SP             172801 non-null  float64
95   \2_PIT_001_PV             172801 non-null  float64
96   \2_PIT_002_PV             172801 non-null  float64
97   \2_PIT_003_PV             172801 non-null  float64
98   \2_SV_101_STATUS          172801 non-null  int64
99   \2_SV_201_STATUS          172801 non-null  int64
100  \2_SV_301_STATUS          172801 non-null  int64
101  \2_SV_401_STATUS          172801 non-null  int64
102  \2_SV_501_STATUS          172801 non-null  int64
103  \2_SV_601_STATUS          172801 non-null  int64
104  \2A_AIT_001_PV            172801 non-null  float64
105  \2A_AIT_002_PV            172801 non-null  float64
106  \2A_AIT_003_PV            172801 non-null  float64
107  \2A_AIT_004_PV            172801 non-null  float64
108  \2B_AIT_001_PV            172801 non-null  float64
109  \2B_AIT_002_PV            172801 non-null  float64
110  \2B_AIT_003_PV            172801 non-null  float64
111  \2B_AIT_004_PV            172801 non-null  float64
112  \3_AIT_001_PV             172801 non-null  int64
113  \3_AIT_002_PV             172801 non-null  float64
```

```
 114  \3_AIT_003_PV              172801 non-null  float64
 115  \3_AIT_004_PV              172801 non-null  float64
 116  \3_AIT_005_PV              172801 non-null  float64
 117  \3_FIT_001_PV              172801 non-null  float64
 118  \3_LS_001_AL               172801 non-null  int64
 119  \3_LT_001_PV               172801 non-null  float64
 120  \3_MV_001_STATUS           172801 non-null  int64
 121  \3_MV_002_STATUS           172801 non-null  int64
 122  \3_MV_003_STATUS           172801 non-null  int64
 123  \3_P_001_STATUS            172801 non-null  int64
 124  \3_P_002_STATUS            172801 non-null  int64
 125  \3_P_003_STATUS            172801 non-null  int64
 126  \3_P_004_STATUS            172801 non-null  int64
 127  \LEAK_DIFF_PRESSURE        172801 non-null  float64
 128  \PLANT_START_STOP_LOG      172801 non-null  int64
 129  \TOTAL_CONS_REQUIRED_FLOW  172801 non-null  float64
dtypes: float64(71), int64(57), object(2)
memory usage: 171.4+ MB
None
           Row        Date             Time  \1_AIT_001_PV  \1_AIT_002_PV  \
0            1   10/9/2017  6:00:00.000 PM        164.210       0.529486
1            2   10/9/2017  6:00:01.000 PM        164.210       0.529486
2            3   10/9/2017  6:00:02.000 PM        164.210       0.529486
3            4   10/9/2017  6:00:03.000 PM        164.210       0.529486
4            5   10/9/2017  6:00:04.000 PM        164.210       0.529486
…          …          …                …              …             …
172796  172797  10/11/2017  5:59:56.000 PM        172.959       0.547483
172797  172798  10/11/2017  5:59:57.000 PM        172.959       0.547483
172798  172799  10/11/2017  5:59:58.000 PM        172.915       0.583479
172799  172800  10/11/2017  5:59:59.000 PM        172.915       0.583479
172800  172801  10/11/2017  6:00:00.000 PM        172.915       0.583479

        \1_AIT_003_PV  \1_AIT_004_PV  \1_AIT_005_PV  \1_FIT_001_PV  \
0             11.9972        482.480       0.331167       0.001273
1             11.9972        482.480       0.331167       0.001273
2             11.9972        482.480       0.331167       0.001273
3             11.9972        482.480       0.331167       0.001273
4             11.9972        482.480       0.331167       0.001273
…               …              …              …              …
172796        11.9184        466.034       0.318217       0.001222
172797        11.9184        466.034       0.318217       0.001222
172798        11.9211        466.051       0.318317       0.001260
172799        11.9211        466.051       0.318317       0.001260
172800        11.9211        466.051       0.318317       0.001260

        \1_LS_001_AL  …  \3_MV_001_STATUS  \3_MV_002_STATUS  \
0                  0  …                 1                 1
1                  0  …                 1                 1
```

```
2                  0  …                  1                  1
3                  0  …                  1                  1
4                  0  …                  1                  1
…                …  …                  …                  …
172796             0  …                  1                  1
172797             0  …                  1                  1
172798             0  …                  1                  1
172799             0  …                  1                  1
172800             0  …                  1                  1

        \3_MV_003_STATUS  \3_P_001_STATUS  \3_P_002_STATUS  \3_P_003_STATUS  \
0                      1                 1                 1                 1
1                      1                 1                 1                 1
2                      1                 1                 1                 1
3                      1                 1                 1                 1
4                      1                 1                 1                 1
…                    …                 …                 …                 …
172796                 1                 1                 1                 1
172797                 1                 1                 1                 1
172798                 1                 1                 1                 1
172799                 1                 1                 1                 1
172800                 1                 1                 1                 1

        \3_P_004_STATUS  \LEAK_DIFF_PRESSURE  \PLANT_START_STOP_LOG  \
0                      1              62.6226                      1
1                      1              62.6226                      1
2                      1              62.6226                      1
3                      1              62.6226                      1
4                      1              62.6226                      1
…                    …                    …                    …
172796                 1              59.3546                      1
172797                 1              59.3546                      1
172798                 1              58.8102                      1
172799                 1              58.8102                      1
172800                 1              58.8102                      1

        \TOTAL_CONS_REQUIRED_FLOW
0                            0.39
1                            0.39
2                            0.39
3                            0.39
4                            0.39
…                            …
172796                       0.00
172797                       0.00
172798                       0.00
172799                       0.00
172800                       0.00
```

```
[172801 rows x 130 columns]
```

1.3.3 Data Cleaning

[9]:
```
print(f"Perform Data Cleaning")
# Remove duplicates (if any)
train17_df.drop_duplicates(inplace=True)
test17_df.drop_duplicates(inplace=True)
```

```
Perform Data Cleaning
```

[10]:
```
# Find columns where the count of non-null values is equal to 0
cols_to_drop = train17_df.columns[train17_df.notnull().sum() == 0]

# Drop those columns from the DataFrame
train17_df = train17_df.drop(columns=cols_to_drop)
```

[11]:
```
# Find columns where the count of non-null values is equal to 0
cols_to_drop = test17_df.columns[test17_df.notnull().sum() == 0]

# Drop those columns from the DataFrame
test17_df = test17_df.drop(columns=cols_to_drop)
```

[12]:
```
# Check for null values
pd.set_option('display.max_rows', None)
train17_df.isnull().sum()
pd.reset_option('display.max_rows')
```

[13]:
```
# Check for null values
pd.set_option('display.max_rows', None)
test17_df.isnull().sum()
pd.reset_option('display.max_rows')
```

[14]:
```
# Remove null values
train17_df = train17_df.ffill().bfill()
test17_df = test17_df.ffill().bfill()
```

[15]:
```
# Check for null values after remoe
train17_df.isnull().sum().sum()
test17_df.isnull().sum().sum()
```

[15]:
```
np.int64(0)
```

1.4 Classify data columns into Meta Group, Sensor Group or Actuator Group

[16]:
```
print(f"Classify the data columns into MetaGroup, Sensor Group and Actuator␣
 ↪Group - 14days")
# ========================================
# EXPLICIT COLUMN GROUPING
# ========================================
all_cols = train17_df.columns.tolist()

metadata_cols = []
actuator_ctrl_cols = []
sensor_cols = []

for col in all_cols:
    col_upper = col.upper()

    # Identify Metadata columns
    if any(k in col_upper for k in ["ROW", "DATE", "TIME"]):
        metadata_cols.append(col)

    # Identify Continuous Sensor Process Values (_PV and direct flow/pressure␣
 ↪metrics)
    elif "_PV" in col_upper or any(
        k in col_upper for k in ["LEAK_DIFF_PRESSURE",␣
 ↪"TOTAL_CONS_REQUIRED_FLOW"]
    ):
        sensor_cols.append(col)

    # Identify Actuators, Alarms, Controller Outputs, and Setpoints␣
 ↪(non-continuous sensors)
    elif any(
        k in col_upper
        for k in [
            "STATUS",
            "_AL",
            "_AH",
            "_CO",
            "_SP",
            "_SPEED",
            "PLANT_START_STOP_LOG",
        ]
    ):
        actuator_ctrl_cols.append(col)

    else:
        # Fallback safety: treat anything else as actuator/control if not␣
 ↪explicitly a sensor
```

```
        actuator_ctrl_cols.append(col)

print(f"\nGroup Breakdown:")
print(f" - Metadata Columns: {len(metadata_cols)}")
print(f" - Actuator Columns: {len(actuator_ctrl_cols)}")
print(f" - Sensor Columns (Target for K-S): {len(sensor_cols)}")
```

```
Classify the data columns into MetaGroup, Sensor Group and Actuator Group -
14days

Group Breakdown:
 - Metadata Columns: 3
 - Actuator Columns: 77
 - Sensor Columns (Target for K-S): 46
```

[17]:
```
print(f"Classify the data columns into MetaGroup, Sensor Group and Actuator␣
 ↪Group - attackdata")
# ========================================
# EXPLICIT COLUMN GROUPING
# ========================================
tall_cols = test17_df.columns.tolist()

tmetadata_cols = []
tactuator_ctrl_cols = []
tsensor_cols = []

for col in tall_cols:
    col_upper = col.upper()

    # Identify Metadata columns
    if any(k in col_upper for k in ["ROW", "DATE", "TIME"]):
        tmetadata_cols.append(col)

     # Identify Continuous Sensor Process Values (_PV and direct flow/pressure␣
 ↪metrics)
    elif "_PV" in col_upper or any(
        k in col_upper for k in ["LEAK_DIFF_PRESSURE",␣
 ↪"TOTAL_CONS_REQUIRED_FLOW"]
    ):
        tsensor_cols.append(col)

    # Identify Actuators, Alarms, Controller Outputs, and Setpoints␣
 ↪(non-continuous sensors)
    elif any(
        k in col_upper
        for k in [
            "STATUS",
```

```
            "_AL",
            "_AH",
            "_CO",
            "_SP",
            "_SPEED",
            "PLANT_START_STOP_LOG",
        ]
    ):
        tactuator_ctrl_cols.append(col)


    else:
        # Fallback safety: treat anything else as actuator/control if not␣
 ↪explicitly a sensor
        tactuator_ctrl_cols.append(col)

print(f"\nGroup Breakdown:")
print(f" - Metadata Columns: {len(tmetadata_cols)}")
print(f" - Actuator Columns: {len(tactuator_ctrl_cols)}")
print(f" - Sensor Columns (Target for K-S): {len(tsensor_cols)}")
```

```
Classify the data columns into MetaGroup, Sensor Group and Actuator Group -
attackdata

Group Breakdown:
 - Metadata Columns: 3
 - Actuator Columns: 77
 - Sensor Columns (Target for K-S): 46
```

1.5 Normalize the datasets (Sensor Columns Only)

[18]:
```
print(f"Normalize the Dataset (sensor columns only)-before normalization-␣
 ↪14days")
print(train17_df.head())
```

```
Normalize the Dataset (sensor columns only)-before normalization- 14days
   Row       Date            Time  \1_AIT_001_PV  \1_AIT_002_PV  \
0    1  9/25/2017  6:00:00.000 PM        171.155       0.619473
1    2  9/25/2017  6:00:01.000 PM        171.155       0.619473
2    3  9/25/2017  6:00:02.000 PM        171.155       0.619473
3    4  9/25/2017  6:00:03.000 PM        171.155       0.607477
4    5  9/25/2017  6:00:04.000 PM        171.155       0.607477

   \1_AIT_003_PV  \1_AIT_004_PV  \1_AIT_005_PV  \1_FIT_001_PV  \1_LS_001_AL  \
0        11.5759        504.645       0.318319       0.001157             0
1        11.5759        504.645       0.318319       0.001157             0
2        11.5759        504.645       0.318319       0.001157             0
```

```
3        11.5725        504.673       0.318438       0.001207              0
4        11.5725        504.673       0.318438       0.001207              0

   …  \3_MV_001_STATUS  \3_MV_002_STATUS  \3_MV_003_STATUS  \3_P_001_STATUS  \
0  …                 1                 1                 1                1
1  …                 1                 1                 1                1
2  …                 1                 1                 1                1
3  …                 1                 1                 1                1
4  …                 1                 1                 1                1

   \3_P_002_STATUS  \3_P_003_STATUS  \3_P_004_STATUS  \LEAK_DIFF_PRESSURE  \
0                1                1                1              67.9651
1                1                1                1              67.9651
2                1                1                1              67.9651
3                1                1                1              67.1948
4                1                1                1              67.1948

   \PLANT_START_STOP_LOG  \TOTAL_CONS_REQUIRED_FLOW
0                      1                       0.68
1                      1                       0.68
2                      1                       0.68
3                      1                       0.68
4                      1                       0.68

[5 rows x 126 columns]
```

[21]:
```
print(f"Normalize the Dataset (sensor columns only)-after normalization-␣
 ↪14days")
# data preprocessing
from sklearn.preprocessing import StandardScaler

# Standardizing the data (data normalization)
scaler = StandardScaler()
scaler = scaler.fit(train17_df[sensor_cols])
train17_df[sensor_cols] = scaler.transform(train17_df[sensor_cols])
print(train17_df.head())
```

```
Normalize the Dataset (sensor columns only)-after normalization- 14days
   Row       Date            Time  \1_AIT_001_PV  \1_AIT_002_PV  \
0    1  9/25/2017  6:00:00.000 PM       0.268563      -0.035593
1    2  9/25/2017  6:00:01.000 PM       0.268563      -0.035593
2    3  9/25/2017  6:00:02.000 PM       0.268563      -0.035593
3    4  9/25/2017  6:00:03.000 PM       0.268563      -0.238219
4    5  9/25/2017  6:00:04.000 PM       0.268563      -0.238219

   \1_AIT_003_PV  \1_AIT_004_PV  \1_AIT_005_PV  \1_FIT_001_PV  \1_LS_001_AL  \
0      -0.921077       0.721884       0.353323      -0.614145             0
1      -0.921077       0.721884       0.353323      -0.614145             0
```

```
2      -0.921077       0.721884       0.353323      -0.614145             0
3      -0.938816       0.723084       0.355775      -0.614087             0
4      -0.938816       0.723084       0.355775      -0.614087             0

   …  \3_MV_001_STATUS  \3_MV_002_STATUS  \3_MV_003_STATUS  \3_P_001_STATUS  \
0  …                 1                 1                 1                1
1  …                 1                 1                 1                1
2  …                 1                 1                 1                1
3  …                 1                 1                 1                1
4  …                 1                 1                 1                1

   \3_P_002_STATUS  \3_P_003_STATUS  \3_P_004_STATUS  \LEAK_DIFF_PRESSURE  \
0                1                1                1             0.841823
1                1                1                1             0.841823
2                1                1                1             0.841823
3                1                1                1             0.710596
4                1                1                1             0.710596

   \PLANT_START_STOP_LOG  \TOTAL_CONS_REQUIRED_FLOW
0                      1                   0.292942
1                      1                   0.292942
2                      1                   0.292942
3                      1                   0.292942
4                      1                   0.292942

[5 rows x 126 columns]
```

[22]:
```
print(f"Normalize the Dataset (sensor columns only)-before normalization-␣
  ↪attackdata")
print(test17_df.head())
```

```
Normalize the Dataset (sensor columns only)-before normalization- attackdata
   Row       Date            Time  \1_AIT_001_PV  \1_AIT_002_PV  \
0    1  10/9/2017  6:00:00.000 PM         164.21       0.529486
1    2  10/9/2017  6:00:01.000 PM         164.21       0.529486
2    3  10/9/2017  6:00:02.000 PM         164.21       0.529486
3    4  10/9/2017  6:00:03.000 PM         164.21       0.529486
4    5  10/9/2017  6:00:04.000 PM         164.21       0.529486

   \1_AIT_003_PV  \1_AIT_004_PV  \1_AIT_005_PV  \1_FIT_001_PV  \1_LS_001_AL  \
0        11.9972         482.48       0.331167       0.001273             0
1        11.9972         482.48       0.331167       0.001273             0
2        11.9972         482.48       0.331167       0.001273             0
3        11.9972         482.48       0.331167       0.001273             0
4        11.9972         482.48       0.331167       0.001273             0

   …  \3_MV_001_STATUS  \3_MV_002_STATUS  \3_MV_003_STATUS  \3_P_001_STATUS  \
0  …                 1                 1                 1                1
```

```
1  …                 1                 1                 1                 1
2  …                 1                 1                 1                 1
3  …                 1                 1                 1                 1
4  …                 1                 1                 1                 1

   \3_P_002_STATUS  \3_P_003_STATUS  \3_P_004_STATUS  \LEAK_DIFF_PRESSURE  \
0                1                 1                 1              62.6226
1                1                 1                 1              62.6226
2                1                 1                 1              62.6226
3                1                 1                 1              62.6226
4                1                 1                 1              62.6226

   \PLANT_START_STOP_LOG  \TOTAL_CONS_REQUIRED_FLOW
0                      1                       0.39
1                      1                       0.39
2                      1                       0.39
3                      1                       0.39
4                      1                       0.39

[5 rows x 126 columns]
```

[23]:
```
print(f"Normalize the Dataset (sensor columns only)-after normalization-␣
 ↪attackdata")
# data preprocessing
# from sklearn.preprocessing import StandardScaler

# Standardizing the data (data normalization)
scaler = StandardScaler()
scaler = scaler.fit(test17_df[tsensor_cols])
test17_df[tsensor_cols] = scaler.transform(test17_df[tsensor_cols])
print(test17_df.head())
```

```
Normalize the Dataset (sensor columns only)-after normalization- attackdata
   Row       Date            Time  \1_AIT_001_PV  \1_AIT_002_PV  \
0    1  10/9/2017  6:00:00.000 PM      -0.642796      -0.339731
1    2  10/9/2017  6:00:01.000 PM      -0.642796      -0.339731
2    3  10/9/2017  6:00:02.000 PM      -0.642796      -0.339731
3    4  10/9/2017  6:00:03.000 PM      -0.642796      -0.339731
4    5  10/9/2017  6:00:04.000 PM      -0.642796      -0.339731

   \1_AIT_003_PV  \1_AIT_004_PV  \1_AIT_005_PV  \1_FIT_001_PV  \1_LS_001_AL  \
0       0.494158       1.521308        1.49528      -0.627893             0
1       0.494158       1.521308        1.49528      -0.627893             0
2       0.494158       1.521308        1.49528      -0.627893             0
3       0.494158       1.521308        1.49528      -0.627893             0
4       0.494158       1.521308        1.49528      -0.627893             0

   …  \3_MV_001_STATUS  \3_MV_002_STATUS  \3_MV_003_STATUS  \3_P_001_STATUS  \
```

```
0  …                 1                 1                 1                 1
1  …                 1                 1                 1                 1
2  …                 1                 1                 1                 1
3  …                 1                 1                 1                 1
4  …                 1                 1                 1                 1

   \3_P_002_STATUS  \3_P_003_STATUS  \3_P_004_STATUS  \LEAK_DIFF_PRESSURE  \
0                1                 1                 1            -0.013872
1                1                 1                 1            -0.013872
2                1                 1                 1            -0.013872
3                1                 1                 1            -0.013872
4                1                 1                 1            -0.013872

   \PLANT_START_STOP_LOG  \TOTAL_CONS_REQUIRED_FLOW
0                      1                  -0.353392
1                      1                  -0.353392
2                      1                  -0.353392
3                      1                  -0.353392
4                      1                  -0.353392

[5 rows x 126 columns]
```

1.6 Add new Column Attack Label and Split Attack data from attackdata dataset

[24]:
```
print(f"Add Attack Label and classify attack label based on specified time␣
 ↪ranges - attackdata")
# Clean column names in case of whitespace artifacts
test17_df.columns = test17_df.columns.str.strip()

# ========================================
# PARSE TIMESTAMP USING 'Date' AND 'Time' COLUMNS
# ========================================
# Create a temporary datetime series directly from 'DATE' and 'TIME'
# without adding a permanent 'DateTime' column to test17_df
temp_datetime = pd.to_datetime(
    test17_df["Date"].astype(str).str.strip()
    + " "
    + test17_df["Time"].astype(str).str.strip(),
    format='mixed',
    errors="coerce"
)

# ========================================
# DEFINE ATTACK TIME RANGES
# ========================================
attack_ranges = [
```

```
    ("09/10/2017 19:25:00", "09/10/2017 19:50:16"),
    ("10/10/2017 10:24:10", "10/10/2017 10:34:00"),
    ("10/10/2017 10:55:00", "10/10/2017 11:24:00"),
    ("10/10/2017 11:07:46", "10/10/2017 11:12:15"),
    ("10/10/2017 11:30:40", "10/10/2017 11:44:50"),
    ("10/10/2017 13:39:30", "10/10/2017 13:50:40"),
    ("10/10/2017 14:48:17", "10/10/2017 14:59:55"),
    ("10/10/2017 14:53:44", "10/10/2017 15:00:32"),
    ("10/10/2017 17:40:00", "10/10/2017 17:49:40"),
    ("11/10/2017 10:55:00", "11/10/2017 10:56:27"),
    ("11/10/2017 11:17:54", "11/10/2017 11:31:20"),
    ("11/10/2017 11:36:31", "11/10/2017 11:47:00"),
    ("11/10/2017 11:59:00", "11/10/2017 12:05:00"),
    ("11/10/2017 12:07:30", "11/10/2017 12:10:52"),
    ("11/10/2017 12:16:00", "11/10/2017 12:25:36"),
    ("11/10/2017 15:26:30", "11/10/2017 15:37:00"),
]

# ========================================
# INITIALIZE AND POPULATE ATTACK LABELS
# ========================================
# Initialize attack label column to 1 as default
test17_df["ATTACK_LABEL"] = 1

# Iterate through each attack range and mark matching rows as -1
for start_str, end_str in attack_ranges:
    start_dt = pd.to_datetime(start_str, format="%d/%m/%Y %H:%M:%S")
    end_dt = pd.to_datetime(end_str, format="%d/%m/%Y %H:%M:%S")

    mask = (temp_datetime >= start_dt) & (temp_datetime <= end_dt)
    test17_df.loc[mask, "ATTACK_LABEL"] = -1

# ========================================
# VERIFICATION
# ========================================
print("\nLabel distribution summary:")
print(test17_df["ATTACK_LABEL"].value_counts())
print(f"\nFinal DataFrame shape: {test17_df.shape}")
print(f"Last column name: {test17_df.columns[-1]}")
```

```
Add Attack Label and classify attack label based on specified time ranges -
attackdata

Label distribution summary:
ATTACK_LABEL
 1    162816
-1      9985
```

```
Name: count, dtype: int64

Final DataFrame shape: (172801, 127)
Last column name: ATTACK_LABEL
```

[25]:
```
print(f"Data info of attackdata after adding Attack Label")
test17_df.info(verbose=True, show_counts=True)
print(test17_df)
```

```
Data info of attackdata after adding Attack Label
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 172801 entries, 0 to 172800
Data columns (total 127 columns):
 #    Column                      Non-Null Count   Dtype
---   ------                      --------------   -----
 0    Row                         172801 non-null  int64
 1    Date                        172801 non-null  object
 2    Time                        172801 non-null  object
 3    \1_AIT_001_PV               172801 non-null  float64
 4    \1_AIT_002_PV               172801 non-null  float64
 5    \1_AIT_003_PV               172801 non-null  float64
 6    \1_AIT_004_PV               172801 non-null  float64
 7    \1_AIT_005_PV               172801 non-null  float64
 8    \1_FIT_001_PV               172801 non-null  float64
 9    \1_LS_001_AL                172801 non-null  int64
 10   \1_LS_002_AL                172801 non-null  int64
 11   \1_LT_001_PV                172801 non-null  float64
 12   \1_MV_001_STATUS            172801 non-null  int64
 13   \1_MV_002_STATUS            172801 non-null  int64
 14   \1_MV_003_STATUS            172801 non-null  int64
 15   \1_MV_004_STATUS            172801 non-null  int64
 16   \1_P_001_STATUS             172801 non-null  int64
 17   \1_P_002_STATUS             172801 non-null  int64
 18   \1_P_003_STATUS             172801 non-null  int64
 19   \1_P_004_STATUS             172801 non-null  int64
 20   \1_P_005_STATUS             172801 non-null  int64
 21   \1_P_006_STATUS             172801 non-null  int64
 22   \2_DPIT_001_PV              172801 non-null  float64
 23   \2_FIC_101_CO               172801 non-null  float64
 24   \2_FIC_101_PV               172801 non-null  float64
 25   \2_FIC_101_SP               172801 non-null  float64
 26   \2_FIC_201_CO               172801 non-null  float64
 27   \2_FIC_201_PV               172801 non-null  float64
 28   \2_FIC_201_SP               172801 non-null  float64
 29   \2_FIC_301_CO               172801 non-null  float64
 30   \2_FIC_301_PV               172801 non-null  float64
 31   \2_FIC_301_SP               172801 non-null  float64
 32   \2_FIC_401_CO               172801 non-null  float64
```

```
33   \2_FIC_401_PV            172801 non-null  float64
34   \2_FIC_401_SP            172801 non-null  float64
35   \2_FIC_501_CO            172801 non-null  float64
36   \2_FIC_501_PV            172801 non-null  float64
37   \2_FIC_501_SP            172801 non-null  float64
38   \2_FIC_601_CO            172801 non-null  float64
39   \2_FIC_601_PV            172801 non-null  float64
40   \2_FIC_601_SP            172801 non-null  float64
41   \2_FIT_001_PV            172801 non-null  float64
42   \2_FIT_002_PV            172801 non-null  float64
43   \2_FIT_003_PV            172801 non-null  float64
44   \2_FQ_101_PV             172801 non-null  float64
45   \2_FQ_201_PV             172801 non-null  float64
46   \2_FQ_301_PV             172801 non-null  float64
47   \2_FQ_401_PV             172801 non-null  float64
48   \2_FQ_501_PV             172801 non-null  float64
49   \2_FQ_601_PV             172801 non-null  float64
50   \2_LS_101_AH             172801 non-null  int64
51   \2_LS_101_AL             172801 non-null  int64
52   \2_LS_201_AH             172801 non-null  int64
53   \2_LS_201_AL             172801 non-null  int64
54   \2_LS_301_AH             172801 non-null  int64
55   \2_LS_301_AL             172801 non-null  int64
56   \2_LS_401_AH             172801 non-null  int64
57   \2_LS_401_AL             172801 non-null  int64
58   \2_LS_501_AH             172801 non-null  int64
59   \2_LS_501_AL             172801 non-null  int64
60   \2_LS_601_AH             172801 non-null  int64
61   \2_LS_601_AL             172801 non-null  int64
62   \2_LT_001_PV             172801 non-null  float64
63   \2_LT_002_PV             172801 non-null  float64
64   \2_MCV_007_CO            172801 non-null  int64
65   \2_MCV_101_CO            172801 non-null  float64
66   \2_MCV_201_CO            172801 non-null  float64
67   \2_MCV_301_CO            172801 non-null  float64
68   \2_MCV_401_CO            172801 non-null  float64
69   \2_MCV_501_CO            172801 non-null  float64
70   \2_MCV_601_CO            172801 non-null  float64
71   \2_MV_001_STATUS         172801 non-null  int64
72   \2_MV_002_STATUS         172801 non-null  int64
73   \2_MV_003_STATUS         172801 non-null  int64
74   \2_MV_004_STATUS         172801 non-null  int64
75   \2_MV_005_STATUS         172801 non-null  int64
76   \2_MV_006_STATUS         172801 non-null  int64
77   \2_MV_009_STATUS         172801 non-null  int64
78   \2_MV_101_STATUS         172801 non-null  int64
79   \2_MV_201_STATUS         172801 non-null  int64
80   \2_MV_301_STATUS         172801 non-null  int64
```

```
 81   \2_MV_401_STATUS           172801 non-null  int64
 82   \2_MV_501_STATUS           172801 non-null  int64
 83   \2_MV_601_STATUS           172801 non-null  int64
 84   \2_P_003_SPEED             172801 non-null  float64
 85   \2_P_003_STATUS            172801 non-null  int64
 86   \2_P_004_SPEED             172801 non-null  float64
 87   \2_P_004_STATUS            172801 non-null  int64
 88   \2_PIC_003_CO              172801 non-null  float64
 89   \2_PIC_003_PV              172801 non-null  float64
 90   \2_PIC_003_SP              172801 non-null  float64
 91   \2_PIT_001_PV              172801 non-null  float64
 92   \2_PIT_002_PV              172801 non-null  float64
 93   \2_PIT_003_PV              172801 non-null  float64
 94   \2_SV_101_STATUS           172801 non-null  int64
 95   \2_SV_201_STATUS           172801 non-null  int64
 96   \2_SV_301_STATUS           172801 non-null  int64
 97   \2_SV_401_STATUS           172801 non-null  int64
 98   \2_SV_501_STATUS           172801 non-null  int64
 99   \2_SV_601_STATUS           172801 non-null  int64
 100  \2A_AIT_001_PV             172801 non-null  float64
 101  \2A_AIT_002_PV             172801 non-null  float64
 102  \2A_AIT_003_PV             172801 non-null  float64
 103  \2A_AIT_004_PV             172801 non-null  float64
 104  \2B_AIT_001_PV             172801 non-null  float64
 105  \2B_AIT_002_PV             172801 non-null  float64
 106  \2B_AIT_003_PV             172801 non-null  float64
 107  \2B_AIT_004_PV             172801 non-null  float64
 108  \3_AIT_001_PV              172801 non-null  float64
 109  \3_AIT_002_PV              172801 non-null  float64
 110  \3_AIT_003_PV              172801 non-null  float64
 111  \3_AIT_004_PV              172801 non-null  float64
 112  \3_AIT_005_PV              172801 non-null  float64
 113  \3_FIT_001_PV              172801 non-null  float64
 114  \3_LS_001_AL               172801 non-null  int64
 115  \3_LT_001_PV               172801 non-null  float64
 116  \3_MV_001_STATUS           172801 non-null  int64
 117  \3_MV_002_STATUS           172801 non-null  int64
 118  \3_MV_003_STATUS           172801 non-null  int64
 119  \3_P_001_STATUS            172801 non-null  int64
 120  \3_P_002_STATUS            172801 non-null  int64
 121  \3_P_003_STATUS            172801 non-null  int64
 122  \3_P_004_STATUS            172801 non-null  int64
 123  \LEAK_DIFF_PRESSURE        172801 non-null  float64
 124  \PLANT_START_STOP_LOG      172801 non-null  int64
 125  \TOTAL_CONS_REQUIRED_FLOW  172801 non-null  float64
 126  ATTACK_LABEL               172801 non-null  int64
dtypes: float64(68), int64(57), object(2)
memory usage: 167.4+ MB
```

```
           Row        Date            Time  \1_AIT_001_PV  \1_AIT_002_PV  \
0            1   10/9/2017  6:00:00.000 PM      -0.642796      -0.339731
1            2   10/9/2017  6:00:01.000 PM      -0.642796      -0.339731
2            3   10/9/2017  6:00:02.000 PM      -0.642796      -0.339731
3            4   10/9/2017  6:00:03.000 PM      -0.642796      -0.339731
4            5   10/9/2017  6:00:04.000 PM      -0.642796      -0.339731
…          …         …              …              …              …
172796  172797  10/11/2017  5:59:56.000 PM      -0.174161      -0.288534
172797  172798  10/11/2017  5:59:57.000 PM      -0.174161      -0.288534
172798  172799  10/11/2017  5:59:58.000 PM      -0.176517      -0.186134
172799  172800  10/11/2017  5:59:59.000 PM      -0.176517      -0.186134
172800  172801  10/11/2017  6:00:00.000 PM      -0.176517      -0.186134

        \1_AIT_003_PV  \1_AIT_004_PV  \1_AIT_005_PV  \1_FIT_001_PV  \
0            0.494158       1.521308       1.495280      -0.627893
1            0.494158       1.521308       1.495280      -0.627893
2            0.494158       1.521308       1.495280      -0.627893
3            0.494158       1.521308       1.495280      -0.627893
4            0.494158       1.521308       1.495280      -0.627893
…                …              …              …              …
172796      -0.071880       0.649421       1.153118      -0.627952
172797      -0.071880       0.649421       1.153118      -0.627952
172798      -0.052486       0.650322       1.155760      -0.627908
172799      -0.052486       0.650322       1.155760      -0.627908
172800      -0.052486       0.650322       1.155760      -0.627908

        \1_LS_001_AL  …  \3_MV_002_STATUS  \3_MV_003_STATUS  \
0                  0  …                 1                 1
1                  0  …                 1                 1
2                  0  …                 1                 1
3                  0  …                 1                 1
4                  0  …                 1                 1
…                …  …  …               …                 …
172796             0  …                 1                 1
172797             0  …                 1                 1
172798             0  …                 1                 1
172799             0  …                 1                 1
172800             0  …                 1                 1

        \3_P_001_STATUS  \3_P_002_STATUS  \3_P_003_STATUS  \3_P_004_STATUS  \
0                     1                1                1                 1
1                     1                1                1                 1
2                     1                1                1                 1
3                     1                1                1                 1
4                     1                1                1                 1
…                   …                …                …                 …
172796                1                1                1                 1
172797                1                1                1                 1
```

```
172798                1                1                1                1
172799                1                1                1                1
172800                1                1                1                1

        \LEAK_DIFF_PRESSURE  \PLANT_START_STOP_LOG  \TOTAL_CONS_REQUIRED_FLOW  \
0                 -0.013872                      1                  -0.353392
1                 -0.013872                      1                  -0.353392
2                 -0.013872                      1                  -0.353392
3                 -0.013872                      1                  -0.353392
4                 -0.013872                      1                  -0.353392
…                       …                      …                          …
172796            -0.553200                      1                  -1.201093
172797            -0.553200                      1                  -1.201093
172798            -0.643043                      1                  -1.201093
172799            -0.643043                      1                  -1.201093
172800            -0.643043                      1                  -1.201093

        ATTACK_LABEL
0                  1
1                  1
2                  1
3                  1
4                  1
…                  …
172796             1
172797             1
172798             1
172799             1
172800             1

[172801 rows x 127 columns]
```

[26]:
```python
print(f"Separate the attackdata into 2 datasets: 1 with normal data and 1 with attack data")
# ========================================
# SEPARATE DATASET BY ATTACK LABEL
# ========================================

# Filter rows where ATTACK_LABEL is 1 (Normal data)
test17_normal_df = test17_df[test17_df["ATTACK_LABEL"] == 1].copy()

# Filter rows where ATTACK_LABEL is -1 (Attack data)
test17_attack_df = test17_df[test17_df["ATTACK_LABEL"] == -1].copy()

# ========================================
# VERIFICATION
# ========================================
```

```
print(f"Original DataFrame shape: {test17_df.shape}")
print(f"Normal DataFrame shape (`test17_normal_df`): {test17_normal_df.shape}")
print(f"Attack DataFrame shape (`test17_attack_df`): {test17_attack_df.shape}")
```

```
Separate the attackdata into 2 datasets: 1 with normal data and 1 with attack
data
Original DataFrame shape: (172801, 127)
Normal DataFrame shape (`test17_normal_df`): (162816, 127)
Attack DataFrame shape (`test17_attack_df`): (9985, 127)
```

1.7 Task 1a Compute K-S Statistic

[29]:
```
print(f"Task 1a Compute K-S Statistic excluding data under attack (Sensor␣
 ↪Columns)")
# ========================================
# CUSTOM K-S STATISTIC FUNCTION
# ========================================
def compute_ks_statistic(sample1, sample2):
    """Computes the two-sample Kolmogorov-Smirnov statistic
    """
    s1 = np.sort(sample1)
    s2 = np.sort(sample2)

    data_all = np.unique(np.concatenate([s1, s2]))

    # Compute Empirical Cumulative Distribution Functions (ECDFs)
    ecdf1 = np.searchsorted(s1, data_all, side="right") / len(s1)
    ecdf2 = np.searchsorted(s2, data_all, side="right") / len(s2)

    # Find the maximum absolute vertical distance between ECDFs
    ks_stat = np.max(np.abs(ecdf1 - ecdf2))
    return ks_stat


# ========================================
# ITERATE AND COMPARE SENSOR GROUP COLUMN-BY-COLUMN
# ========================================
ks_results = {}

for col in sensor_cols:
    train_sample = train17_df[col].values
    test_normal_sample = test17_normal_df[col].values

    # Compute K-S statistic between train17 baseline and normal test17 subset
    ks_val = compute_ks_statistic(train_sample, test_normal_sample)
    ks_results[col] = ks_val
```

```python
# Summarize results into a DataFrame
ks_df = pd.DataFrame(
    list(ks_results.items()), columns=["Sensor_Column", "KS_Statistic"]
)
print("\nSensor List from Highest Distribution Shift (K-S Statistic):")
print(ks_df.sort_values(by="KS_Statistic", ascending=False).head(100))
```

```
Task 1a Compute K-S Statistic excluding data under attack (Sensor Columns)

Sensor List from Highest Distribution Shift (K-S Statistic):
                Sensor_Column  KS_Statistic
30              \2A_AIT_002_PV      0.998219
37               \3_AIT_001_PV      0.982258
38               \3_AIT_002_PV      0.852487
16               \2_FIT_003_PV      0.833281
14               \2_FIT_001_PV      0.765988
5                \1_FIT_001_PV      0.746069
28               \2_PIT_003_PV      0.628026
25               \2_PIC_003_PV      0.627600
34              \2B_AIT_002_PV      0.462716
1                \1_AIT_002_PV      0.416215
0                \1_AIT_001_PV      0.397127
15               \2_FIT_002_PV      0.345660
2                \1_AIT_003_PV      0.300564
31              \2A_AIT_003_PV      0.281691
29              \2A_AIT_001_PV      0.276682
35              \2B_AIT_003_PV      0.256802
39               \3_AIT_003_PV      0.238889
10               \2_FIC_301_PV      0.221438
19                \2_FQ_301_PV      0.221383
11               \2_FIC_401_PV      0.204735
20                \2_FQ_401_PV      0.204553
33              \2B_AIT_001_PV      0.195632
42               \3_FIT_001_PV      0.185658
41               \3_AIT_005_PV      0.185370
13               \2_FIC_601_PV      0.182586
22                \2_FQ_601_PV      0.182550
45  \TOTAL_CONS_REQUIRED_FLOW      0.178485
24                \2_LT_002_PV      0.170129
12               \2_FIC_501_PV      0.163877
21                \2_FQ_501_PV      0.163723
23                \2_LT_001_PV      0.158147
26               \2_PIT_001_PV      0.158090
8                \2_FIC_101_PV      0.152976
17                \2_FQ_101_PV      0.152954
6                 \1_LT_001_PV      0.135713
9                \2_FIC_201_PV      0.116943
```

```
18                \2_FQ_201_PV      0.116924
36              \2B_AIT_004_PV      0.115665
43                \3_LT_001_PV      0.088684
4                 \1_AIT_005_PV      0.080304
7               \2_DPIT_001_PV      0.078434
27                \2_PIT_002_PV      0.074293
40                \3_AIT_004_PV      0.066376
3                 \1_AIT_004_PV      0.065079
32              \2A_AIT_004_PV      0.063061
44         \LEAK_DIFF_PRESSURE      0.048736
```

1.8 Task 1b Compute System State and Percentages

[30]:
```
print(f"Task 1b Compute the set of System States and their percentages")
# ========================================
# EXTRACT SYSTEM STATES AS SETS
# ========================================
# Clean missing values to ensure consistent tuple conversion
train_actuators = train17_df[actuator_ctrl_cols].fillna(0)
test_normal_actuators = test17_normal_df[actuator_ctrl_cols].fillna(0)

# Convert each row into a hashable tuple representing the system state at that␣
 ↪time step
train_tuples = [tuple(x) for x in train_actuators.to_numpy()]
test_normal_tuples = [tuple(x) for x in test_normal_actuators.to_numpy()]

set_train = set(train_tuples)
set_test_normal = set(test_normal_tuples)

# Count unique system states
count_train = len(set_train)
count_test_normal = len(set_test_normal)

print(f"\nUnique system states in train17_df - WADI_14days: {count_train}")
print(f"Unique system states in test17_normal_df - WADI_attackdata:␣
 ↪{count_test_normal}")

print(f"\nCompute the Frequencies and percentages of System State, list 20 top␣
 ↪percentages")

# ========================================
# CALCULATE FREQUENCIES AND PERCENTAGES
# ========================================
# Count occurrences of each unique system state configuration
train_counts = Counter(train_tuples)
test_normal_counts = Counter(test_normal_tuples)
```

```
total_train_rows = len(train_tuples)
total_test_rows = len(test_normal_tuples)

# Retrieve top 20 most frequent states (state tuple, absolute frequency count)
top_train_states = train_counts.most_common(20)
top_test_normal_states = test_normal_counts.most_common(20)

# ========================================
# DISPLAY TOP 20 RESULTS
# ========================================
print("\n" + "=" * 80)
print("TOP 20 MOST FREQUENT SYSTEM STATES AND PERCENTAGES IN TRAIN17_DF␣
 ↪(WADI_14days)")
print("=" * 80)
for rank, (state, freq) in enumerate(top_train_states, start=1):
    percentage = (freq / total_train_rows) * 100
    print(f"Rank {rank:2d}: Count = {freq:6d} | Percentage = {percentage:5.
 ↪10f}%")
    # Display the column-value pairs for this system state
    state_dict = dict(zip(actuator_ctrl_cols, state))
    for col, val in state_dict.items():
        print(f"    {col}: {val}")
    print("-" * 80)


print("\n" + "=" * 80)
print("TOP 20 MOST FREQUENT SYSTEM STATES AND PERCENTAGES IN TEST17_NORMAL_DF -␣
 ↪WADI_attackdata exclude attack data")
print("=" * 80)
for rank, (state, freq) in enumerate(top_test_normal_states, start=1):
    percentage = (freq / total_test_rows) * 100
    print(f"Rank {rank:2d}: Count = {freq:6d} | Percentage = {percentage:5.
 ↪10f}%")
    # Display the column-value pairs for this system state
    state_dict = dict(zip(actuator_ctrl_cols, state))
    for col, val in state_dict.items():
        print(f"    {col}: {val}")
    print("-" * 80)
```

```
Task 1b Compute the set of System States and their percentages

Unique system states in train17_df - WADI_14days: 215030
Unique system states in test17_normal_df - WADI_attackdata: 28951

Compute the Frequencies and percentages of System State, list 20 top percentages

================================================================================
```

```
TOP 20 MOST FREQUENT SYSTEM STATES AND PERCENTAGES IN TRAIN17_DF (WADI_14days)
================================================================================
Rank  1: Count =     16 | Percentage = 0.0013227502%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 2.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.19
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.16
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.12
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.1
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.15
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.09
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 0.0
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 0.0
    \2_MCV_601_CO: 0.0
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
```

```
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 2.0
    \2_MV_201_STATUS: 2.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 2.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0116251
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0361308
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  2: Count =     14 | Percentage = 0.0011574065%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.44
```

\2_FIC_201_CO: 100.0
\2_FIC_201_SP: 0.42
\2_FIC_301_CO: 39.7341
\2_FIC_301_SP: 0.24
\2_FIC_401_CO: 100.0
\2_FIC_401_SP: 0.42
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.2
\2_FIC_601_CO: 88.6627
\2_FIC_601_SP: 0.39
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 88.6627
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 39.7341
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 2.0
\2_MV_201_STATUS: 2.0
\2_MV_301_STATUS: 1.0
\2_MV_401_STATUS: 2.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 1.0
\2_P_003_SPEED: -0.0118382
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0300087
\2_P_004_STATUS: 1.0
\2_PIC_003_CO: 50.0
\2_PIC_003_SP: 1.0

```
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  3: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.43
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.24
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.31
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.24
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.23
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.49
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
```

```
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 0.0
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 0.0
    \2_MCV_601_CO: 0.0
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 2.0
    \2_MV_201_STATUS: 2.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 2.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0143133
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0309184
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  4: Count =     12 | Percentage = 0.0009920627%
```

\1_LS_001_AL: 0.0
\1_LS_002_AL: 0.0
\1_MV_001_STATUS: 1.0
\1_MV_002_STATUS: 1.0
\1_MV_003_STATUS: 1.0
\1_MV_004_STATUS: 1.0
\1_P_001_STATUS: 1.0
\1_P_002_STATUS: 1.0
\1_P_003_STATUS: 1.0
\1_P_004_STATUS: 1.0
\1_P_005_STATUS: 1.0
\1_P_006_STATUS: 1.0
\2_FIC_101_CO: 100.0
\2_FIC_101_SP: 0.43
\2_FIC_201_CO: 100.0
\2_FIC_201_SP: 0.24
\2_FIC_301_CO: 100.0
\2_FIC_301_SP: 0.31
\2_FIC_401_CO: 100.0
\2_FIC_401_SP: 0.24
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.23
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.49
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0

```
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 2.0
    \2_MV_201_STATUS: 2.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 2.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0149608
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0296973
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  5: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.43
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.24
    \2_FIC_301_CO: 100.0
```

\2_FIC_301_SP: 0.31
\2_FIC_401_CO: 100.0
\2_FIC_401_SP: 0.24
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.23
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.49
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 2.0
\2_MV_201_STATUS: 2.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 2.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0144936
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0304513
\2_P_004_STATUS: 1.0
\2_PIC_003_CO: 50.0
\2_PIC_003_SP: 1.0
\2_SV_101_STATUS: 1.0
\2_SV_201_STATUS: 1.0
\2_SV_301_STATUS: 1.0

```
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  6: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 2.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.1
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.1
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.1
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.1
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.1
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.1
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
```

```
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 0.0
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 0.0
    \2_MCV_601_CO: 0.0
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 2.0
    \2_MV_201_STATUS: 2.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 2.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.015682
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0295006
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  7: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
```

```
\1_MV_002_STATUS: 1.0
\1_MV_003_STATUS: 1.0
\1_MV_004_STATUS: 2.0
\1_P_001_STATUS: 1.0
\1_P_002_STATUS: 1.0
\1_P_003_STATUS: 1.0
\1_P_004_STATUS: 1.0
\1_P_005_STATUS: 1.0
\1_P_006_STATUS: 1.0
\2_FIC_101_CO: 100.0
\2_FIC_101_SP: 0.1
\2_FIC_201_CO: 100.0
\2_FIC_201_SP: 0.1
\2_FIC_301_CO: 100.0
\2_FIC_301_SP: 0.1
\2_FIC_401_CO: 100.0
\2_FIC_401_SP: 0.1
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.1
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.1
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 2.0
```

```
    \2_MV_201_STATUS: 2.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 2.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0164934
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0285991
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  8: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.26
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.35
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.22
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.25
```

\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.43
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.42
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 2.0
\2_MV_201_STATUS: 2.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 2.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0146903
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0288859
\2_P_004_STATUS: 1.0
\2_PIC_003_CO: 50.0
\2_PIC_003_SP: 1.0
\2_SV_101_STATUS: 1.0
\2_SV_201_STATUS: 1.0
\2_SV_301_STATUS: 1.0
\2_SV_401_STATUS: 1.0
\2_SV_501_STATUS: 1.0
\2_SV_601_STATUS: 1.0

```
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  9: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 2.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.27
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.32
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.17
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.21
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.46
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.39
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
```

```
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 0.0
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 0.0
    \2_MCV_601_CO: 0.0
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 2.0
    \2_MV_201_STATUS: 2.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 2.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0108466
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.03668
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 10: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 2.0
```

\1_P_001_STATUS: 1.0
\1_P_002_STATUS: 1.0
\1_P_003_STATUS: 1.0
\1_P_004_STATUS: 1.0
\1_P_005_STATUS: 1.0
\1_P_006_STATUS: 1.0
\2_FIC_101_CO: 100.0
\2_FIC_101_SP: 0.27
\2_FIC_201_CO: 100.0
\2_FIC_201_SP: 0.2
\2_FIC_301_CO: 100.0
\2_FIC_301_SP: 0.4
\2_FIC_401_CO: 100.0
\2_FIC_401_SP: 0.21
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.26
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.1
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 2.0
\2_MV_201_STATUS: 2.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 2.0

```
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0150919
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0298694
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 11: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 2.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.27
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.2
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.4
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.21
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.26
    \2_FIC_601_CO: 100.0
```

\2_FIC_601_SP: 0.1
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 2.0
\2_MV_201_STATUS: 2.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 2.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0150919
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0293039
\2_P_004_STATUS: 1.0
\2_PIC_003_CO: 50.0
\2_PIC_003_SP: 1.0
\2_SV_101_STATUS: 1.0
\2_SV_201_STATUS: 1.0
\2_SV_301_STATUS: 1.0
\2_SV_401_STATUS: 1.0
\2_SV_501_STATUS: 1.0
\2_SV_601_STATUS: 1.0
\3_LS_001_AL: 1.0
\3_MV_001_STATUS: 1.0
\3_MV_002_STATUS: 1.0

```
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 12: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 2.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.12
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.12
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.13
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.12
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.1
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.11
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 0.0
```

```
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 0.0
    \2_MCV_601_CO: 0.0
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 2.0
    \2_MV_201_STATUS: 2.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 2.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0123546
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0351883
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 13: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 2.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
```

\1_P_004_STATUS: 1.0
\1_P_005_STATUS: 1.0
\1_P_006_STATUS: 1.0
\2_FIC_101_CO: 100.0
\2_FIC_101_SP: 0.12
\2_FIC_201_CO: 100.0
\2_FIC_201_SP: 0.12
\2_FIC_301_CO: 100.0
\2_FIC_301_SP: 0.13
\2_FIC_401_CO: 100.0
\2_FIC_401_SP: 0.12
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.1
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.11
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 2.0
\2_MV_201_STATUS: 2.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 2.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0120595

```
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0352293
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 14: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.47
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.35
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.24
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.51
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.53
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.31
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
```

\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 2.0
\2_MV_201_STATUS: 2.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 2.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0151001
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0292793
\2_P_004_STATUS: 1.0
\2_PIC_003_CO: 50.0
\2_PIC_003_SP: 1.0
\2_SV_101_STATUS: 1.0
\2_SV_201_STATUS: 1.0
\2_SV_301_STATUS: 1.0
\2_SV_401_STATUS: 1.0
\2_SV_501_STATUS: 1.0
\2_SV_601_STATUS: 1.0
\3_LS_001_AL: 1.0
\3_MV_001_STATUS: 1.0
\3_MV_002_STATUS: 1.0
\3_MV_003_STATUS: 1.0
\3_P_001_STATUS: 1.0
\3_P_002_STATUS: 1.0

```
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 15: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.51
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.33
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.5
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.34
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.32
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.26
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 0.0
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 0.0
    \2_MCV_601_CO: 0.0
```

```
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 2.0
    \2_MV_201_STATUS: 2.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 2.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0149854
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.028681
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 16: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
```

\2_FIC_101_CO: 100.0
\2_FIC_101_SP: 0.49
\2_FIC_201_CO: 100.0
\2_FIC_201_SP: 0.43
\2_FIC_301_CO: 100.0
\2_FIC_301_SP: 0.3
\2_FIC_401_CO: 100.0
\2_FIC_401_SP: 0.25
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.25
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.47
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 2.0
\2_MV_201_STATUS: 2.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 2.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0148214
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0299841
\2_P_004_STATUS: 1.0

```
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 17: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.49
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.43
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.3
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.25
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.25
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.47
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
```

\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 2.0
\2_MV_201_STATUS: 2.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 2.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.014805
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0295907
\2_P_004_STATUS: 1.0
\2_PIC_003_CO: 50.0
\2_PIC_003_SP: 1.0
\2_SV_101_STATUS: 1.0
\2_SV_201_STATUS: 1.0
\2_SV_301_STATUS: 1.0
\2_SV_401_STATUS: 1.0
\2_SV_501_STATUS: 1.0
\2_SV_601_STATUS: 1.0
\3_LS_001_AL: 1.0
\3_MV_001_STATUS: 1.0
\3_MV_002_STATUS: 1.0
\3_MV_003_STATUS: 1.0
\3_P_001_STATUS: 1.0
\3_P_002_STATUS: 1.0
\3_P_003_STATUS: 1.0
\3_P_004_STATUS: 1.0
\PLANT_START_STOP_LOG: 1.0

```
--------------------------------------------------------------------------------
Rank 18: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.49
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.43
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.3
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.25
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.25
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.47
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 0.0
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 0.0
    \2_MCV_601_CO: 0.0
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
```

```
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 2.0
    \2_MV_201_STATUS: 2.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 2.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.015346
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0295579
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 19: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.49
    \2_FIC_201_CO: 100.0
```

\2_FIC_201_SP: 0.43
\2_FIC_301_CO: 100.0
\2_FIC_301_SP: 0.3
\2_FIC_401_CO: 100.0
\2_FIC_401_SP: 0.25
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.25
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.47
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 2.0
\2_MV_201_STATUS: 2.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 2.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0153951
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0292301
\2_P_004_STATUS: 1.0
\2_PIC_003_CO: 50.0
\2_PIC_003_SP: 1.0
\2_SV_101_STATUS: 1.0

```
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 20: Count =     12 | Percentage = 0.0009920627%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 2.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.13
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.11
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.1
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.15
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.1
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.09
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
```

```
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 0.0
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 0.0
    \2_MCV_601_CO: 0.0
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 2.0
    \2_MV_201_STATUS: 2.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 2.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0134446
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0291809
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------

================================================================================
```

```
TOP 20 MOST FREQUENT SYSTEM STATES AND PERCENTAGES IN TEST17_NORMAL_DF -
WADI_attackdata exclude attack data
================================================================================
Rank  1: Count =     66 | Percentage = 0.0405365566%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 2.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 26.6792
    \2_FIC_101_SP: 0.12
    \2_FIC_201_CO: 23.2974
    \2_FIC_201_SP: 0.09
    \2_FIC_301_CO: 16.3637
    \2_FIC_301_SP: 0.09
    \2_FIC_401_CO: 22.409
    \2_FIC_401_SP: 0.08
    \2_FIC_501_CO: 14.3711
    \2_FIC_501_SP: 0.09
    \2_FIC_601_CO: 24.7808
    \2_FIC_601_SP: 0.09
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 24.7696
    \2_MCV_201_CO: 14.3692
    \2_MCV_301_CO: 22.4029
    \2_MCV_401_CO: 16.3425
    \2_MCV_501_CO: 23.2737
    \2_MCV_601_CO: 26.6527
    \2_MV_001_STATUS: 1.0
```

```
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 1.0
    \2_MV_201_STATUS: 1.0
    \2_MV_301_STATUS: 1.0
    \2_MV_401_STATUS: 1.0
    \2_MV_501_STATUS: 1.0
    \2_MV_601_STATUS: 1.0
    \2_P_003_SPEED: -0.0129037
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0323936
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  2: Count =     12 | Percentage = 0.0073702830%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 2.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
```

\2_FIC_101_SP: 0.09
\2_FIC_201_CO: 100.0
\2_FIC_201_SP: 0.17
\2_FIC_301_CO: 100.0
\2_FIC_301_SP: 0.09
\2_FIC_401_CO: 100.0
\2_FIC_401_SP: 0.13
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.12
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.09
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 2.0
\2_MV_201_STATUS: 2.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 2.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0115186
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0359342
\2_P_004_STATUS: 1.0
\2_PIC_003_CO: 50.0

```
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  3: Count =     12 | Percentage = 0.0073702830%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 2.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.43
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.24
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.31
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.24
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.23
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.49
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
```

```
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 0.0
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 0.0
    \2_MCV_601_CO: 0.0
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 2.0
    \2_MV_201_STATUS: 2.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 2.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0149034
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0305988
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
```

```
Rank  4: Count =     12 | Percentage = 0.0073702830%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.09
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.1
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.1
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.13
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.11
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.12
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 0.0
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 0.0
    \2_MCV_601_CO: 0.0
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
```

```
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 2.0
    \2_MV_201_STATUS: 2.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 2.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0126742
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0361636
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  5: Count =     11 | Percentage = 0.0067560928%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 2.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.43
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.24
```

\2_FIC_301_CO: 100.0
\2_FIC_301_SP: 0.31
\2_FIC_401_CO: 100.0
\2_FIC_401_SP: 0.24
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.23
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.49
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 2.0
\2_MV_201_STATUS: 2.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 2.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0136003
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0310086
\2_P_004_STATUS: 1.0
\2_PIC_003_CO: 50.0
\2_PIC_003_SP: 1.0
\2_SV_101_STATUS: 1.0
\2_SV_201_STATUS: 1.0

```
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  6: Count =     11 | Percentage = 0.0067560928%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.41
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.21
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.38
    \2_FIC_401_CO: 100.0
    \2_FIC_401_SP: 0.27
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.37
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.42
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
```

```
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 0.0
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 0.0
    \2_MCV_601_CO: 0.0
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 2.0
    \2_MV_201_STATUS: 2.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 2.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0158541
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0306971
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  7: Count =     11 | Percentage = 0.0067560928%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
```

\1_MV_001_STATUS: 1.0
\1_MV_002_STATUS: 1.0
\1_MV_003_STATUS: 1.0
\1_MV_004_STATUS: 1.0
\1_P_001_STATUS: 1.0
\1_P_002_STATUS: 1.0
\1_P_003_STATUS: 1.0
\1_P_004_STATUS: 1.0
\1_P_005_STATUS: 1.0
\1_P_006_STATUS: 1.0
\2_FIC_101_CO: 100.0
\2_FIC_101_SP: 0.1
\2_FIC_201_CO: 100.0
\2_FIC_201_SP: 0.11
\2_FIC_301_CO: 100.0
\2_FIC_301_SP: 0.11
\2_FIC_401_CO: 100.0
\2_FIC_401_SP: 0.12
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.09
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.09
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0

```
    \2_MV_101_STATUS: 2.0
    \2_MV_201_STATUS: 2.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 2.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0117727
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0348441
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  8: Count =     10 | Percentage = 0.0061419025%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 2.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 100.0
    \2_FIC_101_SP: 0.38
    \2_FIC_201_CO: 100.0
    \2_FIC_201_SP: 0.37
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.37
    \2_FIC_401_CO: 100.0
```

\2_FIC_401_SP: 0.17
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.26
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.43
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 0.0
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 0.0
\2_MCV_601_CO: 0.0
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 2.0
\2_MV_201_STATUS: 2.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 2.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0158623
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0292957
\2_P_004_STATUS: 1.0
\2_PIC_003_CO: 50.0
\2_PIC_003_SP: 1.0
\2_SV_101_STATUS: 1.0
\2_SV_201_STATUS: 1.0
\2_SV_301_STATUS: 1.0
\2_SV_401_STATUS: 1.0
\2_SV_501_STATUS: 1.0

```
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank  9: Count =      6 | Percentage = 0.0036851415%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 35.6659
    \2_FIC_101_SP: 0.2
    \2_FIC_201_CO: 68.5648
    \2_FIC_201_SP: 0.1
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.1
    \2_FIC_401_CO: 20.0701
    \2_FIC_401_SP: 0.09
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.11
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.15
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
```

```
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 20.0685
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 68.6983
    \2_MCV_601_CO: 35.6947
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 1.0
    \2_MV_201_STATUS: 1.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 1.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0161573
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0295743
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 10: Count =      6 | Percentage = 0.0036851415%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
```

\1_MV_004_STATUS: 1.0
\1_P_001_STATUS: 1.0
\1_P_002_STATUS: 1.0
\1_P_003_STATUS: 1.0
\1_P_004_STATUS: 1.0
\1_P_005_STATUS: 1.0
\1_P_006_STATUS: 1.0
\2_FIC_101_CO: 34.9805
\2_FIC_101_SP: 0.2
\2_FIC_201_CO: 64.7761
\2_FIC_201_SP: 0.1
\2_FIC_301_CO: 100.0
\2_FIC_301_SP: 0.1
\2_FIC_401_CO: 18.8262
\2_FIC_401_SP: 0.09
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.11
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.15
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 18.8262
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 64.7761
\2_MCV_601_CO: 34.9805
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 1.0
\2_MV_201_STATUS: 1.0
\2_MV_301_STATUS: 2.0

```
    \2_MV_401_STATUS: 1.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0156492
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0291072
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 11: Count =      6 | Percentage = 0.0036851415%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 34.5963
    \2_FIC_101_SP: 0.2
    \2_FIC_201_CO: 62.7999
    \2_FIC_201_SP: 0.1
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.1
    \2_FIC_401_CO: 18.5406
    \2_FIC_401_SP: 0.09
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.11
```

\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.15
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 18.5406
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 62.7999
\2_MCV_601_CO: 34.5963
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 1.0
\2_MV_201_STATUS: 1.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 1.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0158459
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0282139
\2_P_004_STATUS: 1.0
\2_PIC_003_CO: 50.0
\2_PIC_003_SP: 1.0
\2_SV_101_STATUS: 1.0
\2_SV_201_STATUS: 1.0
\2_SV_301_STATUS: 1.0
\2_SV_401_STATUS: 1.0
\2_SV_501_STATUS: 1.0
\2_SV_601_STATUS: 1.0
\3_LS_001_AL: 1.0
\3_MV_001_STATUS: 1.0

```
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 12: Count =      6 | Percentage = 0.0036851415%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 31.939
    \2_FIC_101_SP: 0.13
    \2_FIC_201_CO: 59.0228
    \2_FIC_201_SP: 0.11
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.1
    \2_FIC_401_CO: 18.2414
    \2_FIC_401_SP: 0.15
    \2_FIC_501_CO: 99.954
    \2_FIC_501_SP: 0.1
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.15
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
```

```
    \2_MCV_301_CO: 18.2371
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 59.0936
    \2_MCV_601_CO: 31.9521
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 1.0
    \2_MV_201_STATUS: 1.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 1.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0153787
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0290826
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 13: Count =      6 | Percentage = 0.0036851415%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
```

\1_P_003_STATUS: 1.0
\1_P_004_STATUS: 1.0
\1_P_005_STATUS: 1.0
\1_P_006_STATUS: 1.0
\2_FIC_101_CO: 30.8972
\2_FIC_101_SP: 0.13
\2_FIC_201_CO: 57.1721
\2_FIC_201_SP: 0.11
\2_FIC_301_CO: 100.0
\2_FIC_301_SP: 0.1
\2_FIC_401_CO: 18.2876
\2_FIC_401_SP: 0.15
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.1
\2_FIC_601_CO: 99.4927
\2_FIC_601_SP: 0.09
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 18.2984
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 57.238
\2_MCV_601_CO: 30.9311
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 1.0
\2_MV_201_STATUS: 1.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 1.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0

```
    \2_P_003_SPEED: -0.0150673
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0300333
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 14: Count =      6 | Percentage = 0.0036851415%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 28.9296
    \2_FIC_101_SP: 0.13
    \2_FIC_201_CO: 53.5519
    \2_FIC_201_SP: 0.11
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.1
    \2_FIC_401_CO: 18.4986
    \2_FIC_401_SP: 0.15
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.1
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.09
    \2_LS_101_AH: 0.0
```

\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 18.4986
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 53.5519
\2_MCV_601_CO: 28.9296
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 1.0
\2_MV_201_STATUS: 1.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 1.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0165179
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0292301
\2_P_004_STATUS: 1.0
\2_PIC_003_CO: 50.0
\2_PIC_003_SP: 1.0
\2_SV_101_STATUS: 1.0
\2_SV_201_STATUS: 1.0
\2_SV_301_STATUS: 1.0
\2_SV_401_STATUS: 1.0
\2_SV_501_STATUS: 1.0
\2_SV_601_STATUS: 1.0
\3_LS_001_AL: 1.0
\3_MV_001_STATUS: 1.0
\3_MV_002_STATUS: 1.0
\3_MV_003_STATUS: 1.0
\3_P_001_STATUS: 1.0

```
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 15: Count =      6 | Percentage = 0.0036851415%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 28.0206
    \2_FIC_101_SP: 0.13
    \2_FIC_201_CO: 51.6792
    \2_FIC_201_SP: 0.11
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.1
    \2_FIC_401_CO: 18.5825
    \2_FIC_401_SP: 0.15
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.1
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.09
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 18.5797
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 51.7324
```

```
    \2_MCV_601_CO: 28.0957
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 1.0
    \2_MV_201_STATUS: 1.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 1.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0162065
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0288121
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 16: Count =      6 | Percentage = 0.0036851415%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
```

\1_P_006_STATUS: 1.0
\2_FIC_101_CO: 26.3813
\2_FIC_101_SP: 0.13
\2_FIC_201_CO: 48.0307
\2_FIC_201_SP: 0.11
\2_FIC_301_CO: 100.0
\2_FIC_301_SP: 0.1
\2_FIC_401_CO: 18.7359
\2_FIC_401_SP: 0.15
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.1
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.09
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 18.7359
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 48.0307
\2_MCV_601_CO: 26.3813
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 1.0
\2_MV_201_STATUS: 1.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 1.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0157148
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0288121

```
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 17: Count =      6 | Percentage = 0.0036851415%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 24.9018
    \2_FIC_101_SP: 0.13
    \2_FIC_201_CO: 44.423
    \2_FIC_201_SP: 0.11
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.1
    \2_FIC_401_CO: 18.8671
    \2_FIC_401_SP: 0.15
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.1
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.09
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
```

\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 18.8352
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 44.5337
\2_MCV_601_CO: 24.9447
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 1.0
\2_MV_201_STATUS: 1.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 1.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0161491
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0276484
\2_P_004_STATUS: 1.0
\2_PIC_003_CO: 50.0
\2_PIC_003_SP: 1.0
\2_SV_101_STATUS: 1.0
\2_SV_201_STATUS: 1.0
\2_SV_301_STATUS: 1.0
\2_SV_401_STATUS: 1.0
\2_SV_501_STATUS: 1.0
\2_SV_601_STATUS: 1.0
\3_LS_001_AL: 1.0
\3_MV_001_STATUS: 1.0
\3_MV_002_STATUS: 1.0
\3_MV_003_STATUS: 1.0
\3_P_001_STATUS: 1.0
\3_P_002_STATUS: 1.0
\3_P_003_STATUS: 1.0
\3_P_004_STATUS: 1.0

```
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 18: Count =      6 | Percentage = 0.0036851415%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 24.3428
    \2_FIC_101_SP: 0.13
    \2_FIC_201_CO: 42.6178
    \2_FIC_201_SP: 0.11
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.1
    \2_FIC_401_CO: 18.9154
    \2_FIC_401_SP: 0.15
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.1
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.09
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 18.9146
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 42.7154
    \2_MCV_601_CO: 24.3281
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
```

```
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 1.0
    \2_MV_201_STATUS: 1.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 1.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.016354
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0293367
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 19: Count =      6 | Percentage = 0.0036851415%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 23.2262
    \2_FIC_101_SP: 0.13
```

\2_FIC_201_CO: 39.2335
\2_FIC_201_SP: 0.11
\2_FIC_301_CO: 100.0
\2_FIC_301_SP: 0.1
\2_FIC_401_CO: 18.9971
\2_FIC_401_SP: 0.15
\2_FIC_501_CO: 100.0
\2_FIC_501_SP: 0.1
\2_FIC_601_CO: 100.0
\2_FIC_601_SP: 0.09
\2_LS_101_AH: 0.0
\2_LS_101_AL: 0.0
\2_LS_201_AH: 0.0
\2_LS_201_AL: 0.0
\2_LS_301_AH: 0.0
\2_LS_301_AL: 0.0
\2_LS_401_AH: 0.0
\2_LS_401_AL: 0.0
\2_LS_501_AH: 0.0
\2_LS_501_AL: 0.0
\2_LS_601_AH: 0.0
\2_LS_601_AL: 0.0
\2_MCV_007_CO: 0.0
\2_MCV_101_CO: 0.0
\2_MCV_201_CO: 0.0
\2_MCV_301_CO: 18.9971
\2_MCV_401_CO: 0.0
\2_MCV_501_CO: 39.2335
\2_MCV_601_CO: 23.2262
\2_MV_001_STATUS: 1.0
\2_MV_002_STATUS: 1.0
\2_MV_003_STATUS: 1.0
\2_MV_004_STATUS: 2.0
\2_MV_005_STATUS: 2.0
\2_MV_006_STATUS: 1.0
\2_MV_009_STATUS: 2.0
\2_MV_101_STATUS: 1.0
\2_MV_201_STATUS: 1.0
\2_MV_301_STATUS: 2.0
\2_MV_401_STATUS: 1.0
\2_MV_501_STATUS: 2.0
\2_MV_601_STATUS: 2.0
\2_P_003_SPEED: -0.0153705
\2_P_003_STATUS: 1.0
\2_P_004_SPEED: 0.0280254
\2_P_004_STATUS: 1.0
\2_PIC_003_CO: 50.0
\2_PIC_003_SP: 1.0

```
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
Rank 20: Count =      6 | Percentage = 0.0036851415%
    \1_LS_001_AL: 0.0
    \1_LS_002_AL: 0.0
    \1_MV_001_STATUS: 1.0
    \1_MV_002_STATUS: 1.0
    \1_MV_003_STATUS: 1.0
    \1_MV_004_STATUS: 1.0
    \1_P_001_STATUS: 1.0
    \1_P_002_STATUS: 1.0
    \1_P_003_STATUS: 1.0
    \1_P_004_STATUS: 1.0
    \1_P_005_STATUS: 1.0
    \1_P_006_STATUS: 1.0
    \2_FIC_101_CO: 22.336
    \2_FIC_101_SP: 0.13
    \2_FIC_201_CO: 36.0177
    \2_FIC_201_SP: 0.11
    \2_FIC_301_CO: 100.0
    \2_FIC_301_SP: 0.1
    \2_FIC_401_CO: 19.0056
    \2_FIC_401_SP: 0.15
    \2_FIC_501_CO: 100.0
    \2_FIC_501_SP: 0.1
    \2_FIC_601_CO: 100.0
    \2_FIC_601_SP: 0.09
    \2_LS_101_AH: 0.0
    \2_LS_101_AL: 0.0
    \2_LS_201_AH: 0.0
    \2_LS_201_AL: 0.0
    \2_LS_301_AH: 0.0
    \2_LS_301_AL: 0.0
    \2_LS_401_AH: 0.0
```

```
    \2_LS_401_AL: 0.0
    \2_LS_501_AH: 0.0
    \2_LS_501_AL: 0.0
    \2_LS_601_AH: 0.0
    \2_LS_601_AL: 0.0
    \2_MCV_007_CO: 0.0
    \2_MCV_101_CO: 0.0
    \2_MCV_201_CO: 0.0
    \2_MCV_301_CO: 19.0233
    \2_MCV_401_CO: 0.0
    \2_MCV_501_CO: 36.0811
    \2_MCV_601_CO: 22.386
    \2_MV_001_STATUS: 1.0
    \2_MV_002_STATUS: 1.0
    \2_MV_003_STATUS: 1.0
    \2_MV_004_STATUS: 2.0
    \2_MV_005_STATUS: 2.0
    \2_MV_006_STATUS: 1.0
    \2_MV_009_STATUS: 2.0
    \2_MV_101_STATUS: 1.0
    \2_MV_201_STATUS: 1.0
    \2_MV_301_STATUS: 2.0
    \2_MV_401_STATUS: 1.0
    \2_MV_501_STATUS: 2.0
    \2_MV_601_STATUS: 2.0
    \2_P_003_SPEED: -0.0163622
    \2_P_003_STATUS: 1.0
    \2_P_004_SPEED: 0.0295416
    \2_P_004_STATUS: 1.0
    \2_PIC_003_CO: 50.0
    \2_PIC_003_SP: 1.0
    \2_SV_101_STATUS: 1.0
    \2_SV_201_STATUS: 1.0
    \2_SV_301_STATUS: 1.0
    \2_SV_401_STATUS: 1.0
    \2_SV_501_STATUS: 1.0
    \2_SV_601_STATUS: 1.0
    \3_LS_001_AL: 1.0
    \3_MV_001_STATUS: 1.0
    \3_MV_002_STATUS: 1.0
    \3_MV_003_STATUS: 1.0
    \3_P_001_STATUS: 1.0
    \3_P_002_STATUS: 1.0
    \3_P_003_STATUS: 1.0
    \3_P_004_STATUS: 1.0
    \PLANT_START_STOP_LOG: 1.0
--------------------------------------------------------------------------------
```